

HA TINH, Viet Nam

WMO: 488460

Lat: 18.350N

Lon: 105.900E

Elev: 10

StdP: 14.69

Time Zone: 7.00 (E07)

Period: 96-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	52.7	54.3	46.5	46.8	56.5	49.0	51.5	57.0	11.9	62.1	10.0	63.3	3.7	N/A	0.260

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
6	10.2	97.2	81.4	95.6	81.1	94.0	80.7	83.4	92.9	82.4	91.9	81.7	90.8	5.2	N/A

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
81.0	161.0	88.6	80.1	156.2	87.4	79.3	152.3	86.5	47.5	93.4	46.4	92.4	45.6	91.1	88.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 11.1	9.4	7.9	DB	51.1	100.8	1.3	1.7	50.2	102.1	49.4	103.1	48.7	104.0	47.7	105.2	(4)
(5)			WB	48.5	85.3	2.2	1.4	46.9	86.3	45.7	87.2	44.4	88.0	42.8	89.0	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	76.5	63.8	66.2	70.2	77.7	83.3	86.9	86.6	84.8	81.5	77.5	72.6	66.2	(6)	
	DBStd	9.37	5.58	6.72	6.31	5.70	4.91	3.75	3.02	2.77	3.20	3.57	4.93	5.01	(7)	
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0	(8)	
	HDD65	227	93	61	23	0	0	0	0	0	0	0	5	45	(9)	
	CDD50	9673	428	454	625	830	1033	1107	1135	1079	946	854	679	503	(10)	
	CDD65	4425	56	95	183	380	568	657	670	614	496	389	234	83	(11)	
	CDH74	48273	75	262	865	3484	7084	9276	9365	7979	5479	3083	1165	156	(12)	
	CDH80	21050	6	30	187	1200	3293	5079	4961	3672	1876	594	148	4	(13)	
(14) Wind	WSAvg	2.9	2.9	2.3	2.4	2.5	2.8	3.1	3.0	2.8	2.9	3.6	3.5	3.4	(14)	
(15) Precipitation	PrecAvg	94.5	2.6	1.8	2.4	2.8	7.1	4.6	4.4	9.6	18.7	24.7	8.3	4.9	(15)	
	PrecMax	132.1	5.7	3.3	5.7	7.4	18.5	15.7	11.9	26.2	39.6	51.2	19.0	15.0	(16)	
	PrecMin	63.2	0.0	0.3	0.5	0.4	2.0	0.3	1.1	1.9	6.0	5.0	1.0	0.3	(17)	
	PrecStd	17.5	1.5	0.9	1.2	1.7	3.8	3.8	2.8	5.8	8.5	13.0	5.0	3.3	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	78.7	84.4	91.7	97.7	98.9	98.9	99.1	97.0	93.5	88.7	85.4	80.6	(19)	
		MCWB	72.8	75.9	76.0	81.5	82.2	81.4	81.7	81.2	81.3	79.1	77.9	75.7	(20)	
	2%	DB	75.9	78.9	84.3	92.6	96.2	96.9	96.3	94.0	91.3	86.3	83.5	77.7	(21)	
		MCWB	71.3	73.7	76.1	80.6	81.8	81.0	81.5	80.6	80.3	77.9	76.6	73.3	(22)	
	5%	DB	73.9	76.9	80.5	89.3	94.2	95.2	94.6	92.3	89.3	84.8	81.7	75.8	(23)	
		MCWB	70.0	72.6	75.1	79.5	81.3	80.7	81.1	80.0	79.6	77.1	75.7	71.8	(24)	
	10%	DB	71.9	75.2	78.1	86.0	91.6	93.5	92.9	90.8	87.4	83.3	79.8	73.7	(25)	
		MCWB	68.6	71.5	73.9	78.4	80.5	80.3	80.5	79.6	78.9	76.6	74.8	70.2	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	73.9	76.5	78.2	83.5	85.3	83.7	83.8	83.7	82.7	81.0	79.1	76.8	(27)	
		MCDB	76.9	82.5	85.3	93.3	95.6	93.2	93.3	91.7	90.4	86.6	83.7	79.3	(28)	
	2%	WB	72.3	74.4	77.0	81.5	83.3	82.6	82.8	81.9	81.3	79.4	77.4	74.1	(29)	
		MCDB	75.1	77.8	82.3	90.0	93.3	92.7	92.6	89.6	88.3	83.9	81.7	77.0	(30)	
	5%	WB	70.5	73.1	75.7	80.1	82.0	81.7	81.9	81.1	80.4	78.3	76.4	72.2	(31)	
		MCDB	73.2	76.3	80.3	87.1	91.0	91.9	91.6	88.8	86.8	82.4	80.5	75.0	(32)	
	10%	WB	69.0	71.9	74.3	79.0	81.0	81.0	81.3	80.5	79.6	77.6	75.3	70.6	(33)	
		MCDB	71.6	74.9	78.0	85.3	89.3	91.0	90.6	88.2	85.5	81.5	79.3	73.4	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	5.9	5.9	7.0	8.5	10.2	10.2	10.1	9.8	8.5	7.4	7.0	6.4	(35)	
		MCDBR	8.6	8.3	11.1	14.6	13.5	12.7	12.9	12.5	11.7	9.9	9.4	8.3	(36)	
	5% WB	MCWBR	4.5	4.4	4.8	6.1	5.2	4.7	4.7	4.6	4.2	3.9	4.2	4.4	(37)	
		MCWBR	7.8	8.2	10.3	12.0	11.9	11.4	11.3	11.1	10.4	8.6	8.6	7.5	(38)	
(40) Clear-Sky Solar Irradiance	taub		0.578	0.611	0.717	0.667	0.526	0.469	0.448	0.483	0.553	0.644	0.593	0.626	(40)	
		taud	1.832	1.801	1.625	1.763	2.152	2.325	2.378	2.267	2.043	1.777	1.851	1.738	(41)	
	Ebn at Noon		223	225	207	220	250	262	268	261	242	214	218	205	(42)	
		Edh at Noon	63	68	84	74	49	41	39	44	54	69	61	68	(43)	
(44) All-Sky Solar Radiation	RadAvg		836	1031	1128	1450	1653	1649	1571	1494	1328	1125	980	768	(44)	
	RadStd		147	155	112	129	144	120	191	110	152	155	133	124	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (5 neighbors)		+0.59	N/A	N/A	+0.79	+0.36	+0.29	N/A	N/A	N/A	N/A

Nomenclature: See separate page